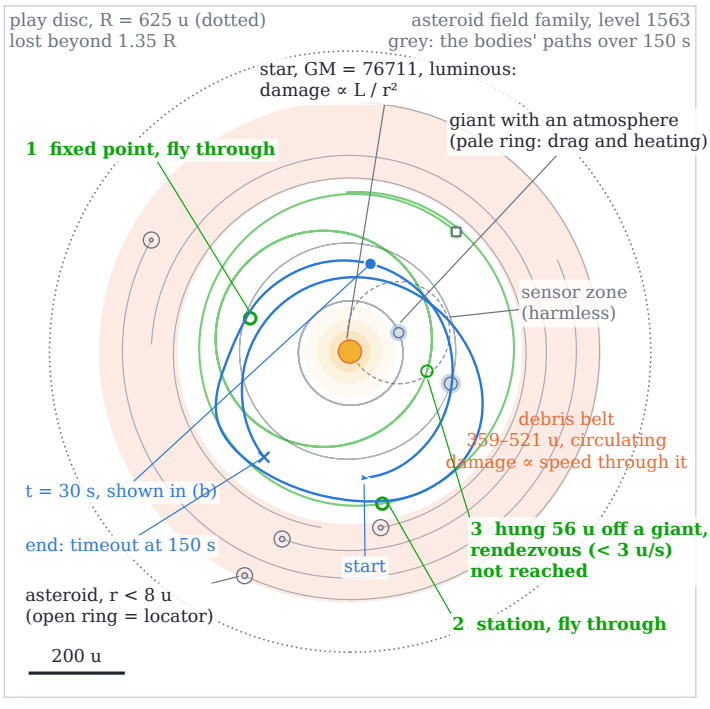
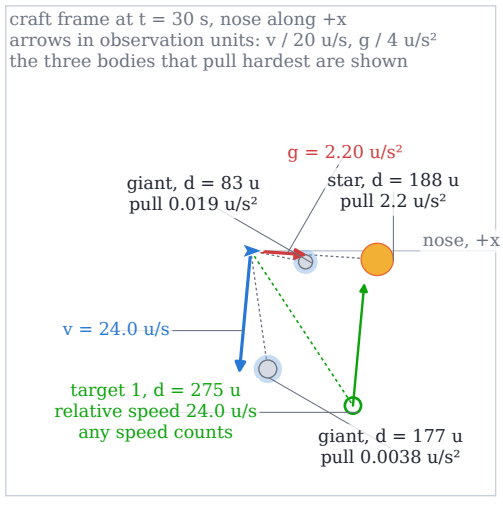
## (a) the level, and one flight

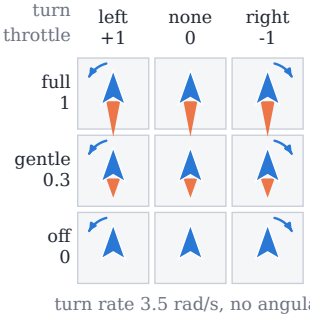



## (b) what the policy observes




**one body token (24 numbers) per body, up to 32:**
relative position, direction, distance, relative velocity, its pull ÷ the engine, GM, radius, atmosphere, luminosity, clearance, is-a-target, kind
**one zone token (15) per zone, up to 12:**
where the band is, its radii, signed distance to it, kind, strength, spin
**self block (109):**
v, g, the current target's offset, distance and relative velocity; fuel, Δv left, Δv left ÷ Δv the tour still needs, engine, health, time left, targets pending, order-free, damage rate, exposure, distance ÷ R, the objective weights, 4 field-readout slots (Sec. 3); 13 per target slot (×6)
tokens are pooled by attention with the self block as the query, so nothing depends on how many bodies there are or their order

## (c) the nine actions




turn rate 3.5 rad/s, no angular inertia; thrust = throttle × engine acceleration; an action is held for one tick (67 ms)

## (d) goal kinds

**fly-through**



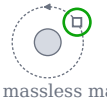
pass within 12 u, at any speed

**rendezvous**



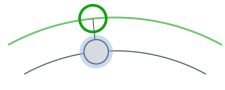

pass within 12 u, and |v craft − v target| < 3 u/s

**station**



a massless marker on its own orbit; fly through or rendezvous

**attached to a body**



rides a planet or moon at a fixed offset; fly through or rendezvous

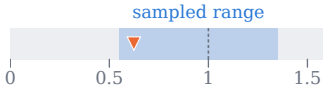
a task is a tour of 1 to 6 such targets, each a ring of radius 12 u, visited in sequence or, when the task says so, in any order

## (e) loadout

engine a ∈ [0.6, 3] u/s², log-uniform

tank f ∈ [3, 60] s of full burn

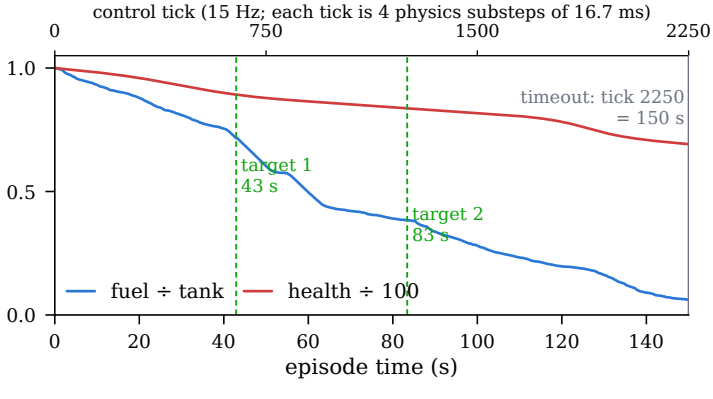
Δv = a · f, with the tank sized so that Δv ÷ a direct-flight estimate is uniform in [0.55, 1.35], then clipped to the tank limits




Δv ÷ direct-flight estimate

this task (▾): a = 0.94 u/s², f = 60 s, Δv = 57 u/s against ≈ 91 u/s for a direct flight, a ratio of 0.62

## (f) the episode clock




**units**

u game length unit (a level is 1000–3000 u across)
s game second; physics at 60 Hz, the pilot at 15 Hz
G = 1, so a body's mass is its GM, in u³/s²; a massless body (station, tracer) is carried by gravity, exerts none
speeds in u/s, accelerations in u/s²; health 0–100, damage in /s

## (g) how an episode ends, and units

**how an episode ends (checked every substep)**

| | |
|---|---|
| **arrived** | every target reached; the only success |
| **crashed** | inside a solid body's radius |
| **destroyed** | health ≤ 0: star radiation, atmospheric heating, debris, radiation belts |
| **lost** | farther than 1.35 R from the centre |
| **timeout** | tick 2250 = 150 s, tour unfinished |
| **tank empty** | not an ending: the craft coasts on |